DongYu Weng

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Education

Carleton College Northfield, MN
Bachelor of Arts in Physics and Computer Science *Expected June 2028*

Research

Integrated Optofluidic Devices Sep. 2026 – Present
Carleton College (Prof. Martha-Elizabeth Baylor) *Northfield, MN*

- Characterized photopolymer resins used to fabricate integrated optofluidic devices that combined optical waveguides and fluidic channels on a single chip.
- Built apparatus to measure resin properties across formulations, including refractive index, optical loss per unit length, and stiffness.
- Worked toward a reference table of materials so engineers could choose the right resin for applications ranging from wearables to harsh environments.

Optics Education Research Jan. 2026 – June 2026
Carleton College (Prof. Martha-Elizabeth Baylor) *Northfield, MN*

- Designed and implemented accessible, low-cost optics activities to engage underrepresented communities in foundational physics concepts.
- Assessed student comprehension through activity-based evaluations and refined materials to improve engagement and conceptual understanding.

Experience

Makerspace Manager Sep. 2024 – Present
Carleton College *Northfield, MN*

- Supported students and faculty on projects from soldering to costume fabrication, advising on tools and build methods.
- Led brainstorming sessions and fabricated custom components on request from professors for course and research needs.

Physics Lab Assistant Sep. 2026 – Present
Carleton College *Northfield, MN*

- Provided hands-on instruction in physics lab sections, guiding students through experimental setup, data collection, and analysis.
- Explained the physics behind each experiment and troubleshot equipment so students could work through problems independently.

Engineering Design Associate May 2025 – Aug. 2026
Fuzhou Changhui Light Steel Structure Building Company *Fuzhou, China*

- Converted 2D building schematics into 3D models using CAD software, improving visualization and structural accuracy.
- Collaborated with engineers and designers to ensure compliance with project specifications and safety standards.

Data Analysis Intern Mar. 2024 – June 2024
Office of The State Superintendent of Education *Washington, DC*

- Analyzed chronic absenteeism and truancy trends across DC schools and built Tableau dashboards to present findings to the public.

Barista July 2023 – May 2024
Luna Hall DC *Washington, DC*

- Served as cashier and prepared specialty boba beverages in a fast-paced customer service environment.
- Trained new hires, managed inventory, and handled high-volume rushes and customer issues.

Certifications and Technical Skills

Certifications: Google Cybersecurity, Google Data Analytics, Google IT Automation with Python, Georgetown Cybersecurity
Languages: Python, C/C++, Kotlin, HTML/CSS, Java
Tools: Git, Docker, Fusion 360, Excel, Tableau, VS Code